

Coverage Ledger — feature × test-type × evidence-state schema (§11.4.25/.52/.153/.169)

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Reconciled (§11.4.35, 2026-06-26): two fixes. (1) §1 Axis-B now states the test-type set consistently as **16 bundled families / 18 enumerated codes** (the [10 §2] taxonomy families vs the 18 test_type CHECK codes of §3) — resolving the "sixteen types" / 17-token / 18-code drift. (2) §6's table is relabelled the illustrative **target** state (NOT current): at spec time the §2/§3 CHECK makes any AUTONOMOUS_VERIFIED cell schema-impossible without captured evidence, so the §7 generator seeds every cell at REQUIRED/PENDING.

Volume 8 (Testing & QA) nano-detail specification. This document deepens §6 of the Volume-8 overview [10 §6] into the **canonical schema, generation pipeline, sync model, and release-gate rule** for the HelixVPN **coverage ledger** — the §11.4.25 / §11.4.52 / §11.4.153 mechanical answer to *"is every feature proven, by every warranted §11.4.169 test type, with real captured evidence?"*. It is **SPEC-ONLY**: it fixes the DDL, the cell state machine, the closed evidence-state vocabulary, the coverage_ledger generator contract, the docs_chain sync context, and the §1.1 paired mutation that makes a ledger bluff mechanically impossible — it is not the product. Sources cited inline: the Volume-8 overview [10] (the test-type taxonomy §2, evidence model §3, ledger sketch §6, acceptance gates §7); the obfuscation/DPI spec [obf] (v02-data-plane/obfuscation-and-dpi.md — the transport survival matrix the transport-feature rows trace to); functional requirements [01_FR] (v01-product/functional-requirements.md); nonfunctional requirements [01_NFR] (v01-product/nonfunctional-requirements.md). Unproven facts are flagged UNVERIFIED: per §11.4.6.

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0. What the ledger is, and what it is not

The coverage ledger is a **projection**, not a source of truth about *the work*. It cross-joins two existing sources — the §11.4.93 workable-items DB (every feature/component/item) and the §11.4.169 test-type matrix (every type a feature warrants) — and records, per (feature, test-type) cell, **one evidence-state** plus the **evidence-artifact path** that backs it. Its single job is to make the question "*can this phase ship?*" a mechanical SELECT, never a human judgement ([10 §6]).

It is therefore deliberately **derived and regenerable** (a §11.4.77-class artifact whose regeneration mechanism is the coverage_ledger tool of §7), yet it lives inside the **git-tracked** workable-items DB (docs/.workable_items.db, §11.4.95) because the cell states + evidence paths *are* authoritative QA history, not a disposable build output. The rendered Markdown/HTML/PDF (docs/qa/coverage_ledger.md

- siblings) is the disposable view, regenerated by docs_chain (§11.4.106) from the DB on every QA run.

What it is **not**: it is not a pass/fail test runner (the harnesses of [10 §5] are), not a replacement for the §11.4.108 runtime-signature registry (a cell going AUTONOMOUS_VERIFIED *consumes* a runtime signature; it does not replace it), and not a substitute for the §11.4.40 full-suite retest (it is that retest's **scoreboard**, [10 §6.2]). Honest boundary (§11.4.6): a green ledger proves *every warranted type ran and produced cited evidence* — it does not prove the product is defect-free (un-enumerated features are the §11.4.118 unknown-unknown surface, §11).

1. The two axes: the feature inventory × the §11.4.169 test-type set

Axis A — features. Every user-visible capability + every internal component the spec declares, each a features row keyed F-<SLUG> with a component (helix-core | helix-go | helix-edge | helix-ui | shim-<os>), a §11.4.91 self-contained title (≥40 chars), a §11.4.69 evidence_class, and a phase (P0..P3). The full inventory is enumerated in §5.

Axis B — the test types. The closed §11.4.169 set fixed by [10 §2] — **16 bundled families / 18 enumerated codes:** the [10 §2] taxonomy presents 16 families (UI/UX and CHAL/HQA each render as one family), which expand to the 18 enumerated test_type codes of the §3 CHECK: UNIT INT E2E FA CHAL HQA SEC DDOS STRESS CHAOS CONC RACE MEM BENCH PERF SCALE UI REC. A feature **declares** which subset it warrants (§11.4.169 — the *absence* of a warranted type is never silent; it is an explicit NOT_APPLICABLE: <reason>). A cell exists for every (feature, declared-type) pair; types a feature genuinely does not warrant simply have no cell (e.g. F-IOS-NE-MEM warrants only MEM).

The cross-join features × declared-types is the ledger; each cell carries one **evidence-state** from §2.

2. The closed evidence-state vocabulary (the cell value)

Eight states, a CHECK-enforced closed set (extends the [10 §6.1] enumeration with REQUIRED/PENDING lifecycle states made explicit):

State	Meaning	Evidence path required?	Release-blocking?
REQUIRED	the feature declares this type; no test authored yet	no	yes (Phase gate)
PENDING	test authored, not yet run on the current artifact	no	yes
AUTONOMOUS_DESIGNED	test authored, RED on the broken artifact (§11.4.115), GREEN not yet captured	no	yes
AUTONOMOUS_VERIFIED	GREEN on a clean deploy (§11.4.108/.139) + evidence path captured (§11.4.69)	YES (schema-enforced)	no
OPERATOR_ATTENDED_ONLY		path optional	yes until promoted

State	Meaning	Evidence path required?	Release-blocking?
	only an operator-driven path exists (§11.4.52)		
SKIP	topology/hardware/credential genuinely absent (§11.4.3 closed reason)	skip_reason required	no
NOT_APPLICABLE	the type does not apply to this feature at this phase (§11.4.6)	skip_reason required	no
FAIL	the test regressed on re-run (§11.4.4 STOP)	evidence of the failure	yes

The load-bearing rule (§11.4.69, mechanically enforced by a schema CHECK in §3): **AUTONOMOUS_VERIFIED is impossible without a non-empty evidence_path** — the database itself rejects a "verified" cell with no captured artifact, so the config-only / absence-of-error PASS (the B1/B2 bluff classes of [10 §0]) cannot be recorded. Likewise SKIP/NOT_APPLICABLE cannot be recorded without a closed-set reason, so a silent omission cannot masquerade as a skip (§11.4.3).

OPERATOR_ATTENDED_ONLY is the §11.4.52 release-blocker state: it is honest (the feature *is* validated, just not autonomously) but it does not scale to the §11.4.156 local-CI-less re-run discipline, so it blocks the tag until promoted to AUTONOMOUS_VERIFIED with a tracked migration item.

3. The schema (DDL, git-tracked SQLite §11.4.95)

```
-- docs/.workable_items.db (git-tracked, §11.4.95) – coverage-
    ledger projection.
-- WAL-checkpointed (PRAGMA wal_checkpoint(TRUNCATE)) before stage
    so .db-wal/.db-shm
-- sidecars stay discardable (§11.4.95). Regenerated by
    coverage_ledger (§7).
```

```
CREATE TABLE IF NOT EXISTS features (
  feature_id      TEXT PRIMARY KEY,           -- 'F-
    AUTHZ-REACH'
  component       TEXT NOT NULL,              --
    helix-core|helix-go|helix-edge|helix-ui|shim-<os>
  title           TEXT NOT NULL CHECK (length(title) >= 40), --
    §11.4.91 self-contained meaning
  evidence_class  TEXT NOT NULL,              --
    §11.4.69 class (network_connectivity, ...)
  fr_ref          TEXT,                       --
    HVPN-FR-NNN / HVPN-NFR-NNN traceability (§v00 RTM)
  phase           TEXT NOT NULL CHECK (phase IN
    ('P0', 'P1', 'P2', 'P3'))
```

```

);

CREATE TABLE IF NOT EXISTS coverage_cells (
    feature_id TEXT NOT NULL REFERENCES features(feature_id) ON
        DELETE CASCADE,
    test_type TEXT NOT NULL CHECK (test_type
        IN -- §11.4.169 closed set [10 §2]
        ('UNIT', 'INT', 'E2E', 'FA', 'CHAL', 'HQA', 'SEC', 'DDOS', 'STRESS', 'CHAOS',
        'CONC', 'RACE', 'MEM', 'BENCH', 'PERF', 'SCALE', 'UI', 'REC')),
    state TEXT NOT NULL CHECK (state IN
        ('REQUIRED', 'PENDING', 'AUTONOMOUS_DESIGNED', 'AUTONOMOUS_VERIFIED',

        'OPERATOR_ATTENDED_ONLY', 'SKIP', 'NOT_APPLICABLE', 'FAIL')),
    skip_reason TEXT, --
        closed set (§11.4.3) when SKIP/NA
    evidence_path TEXT, --
        §11.4.69 captured artifact
    runtime_sig TEXT, --
        §11.4.108 signature id consumed on VERIFIED
    artifact_md5 TEXT, --
        the clean-deploy artifact the evidence is from (§11.4.139)
    last_run_at TEXT, --
        ISO-8601 UTC
    n_iters INTEGER, --
        §11.4.50 determinism count (3 normal / 10 cycle)
    diary_ref INTEGER, --
        §11.4.149 test_diary.diary_id
    challenge_id TEXT, --
        CHAL/HQA: the challenges bank entry id
    PRIMARY KEY (feature_id, test_type),

    -- §11.4.69: a VERIFIED cell MUST cite a captured artifact
    (defeats config-only PASS).
    CHECK (state <> 'AUTONOMOUS_VERIFIED'
        OR (evidence_path IS NOT NULL AND
        length(evidence_path) > 0)),
    -- §11.4.139: a VERIFIED cell MUST record the clean artifact
    its evidence came from.
    CHECK (state <> 'AUTONOMOUS_VERIFIED'
        OR (artifact_md5 IS NOT NULL AND length(artifact_md5)
        > 0)),
    -- §11.4.3: a SKIP/NA cell MUST carry a closed-set reason (no
    silent omission).
    CHECK (state NOT IN ('SKIP', 'NOT_APPLICABLE') OR (skip_reason
        IS NOT NULL))
);

-- The release-blocker set: any phase ships only when this view is
empty for it.
CREATE VIEW IF NOT EXISTS v_coverage_gaps AS
    SELECT f.phase, f.feature_id, c.test_type, c.state,
        c.skip_reason
    FROM features f JOIN coverage_cells c USING (feature_id)
    WHERE c.state IN
        ('REQUIRED', 'PENDING', 'AUTONOMOUS_DESIGNED', 'OPERATOR_ATTENDED_ONLY', 'FAIL

```

-- Per-feature roll-up for the rendered ledger + §11.4.132 risk-ordering input.

```
CREATE VIEW IF NOT EXISTS v_feature_rollup AS
  SELECT f.feature_id, f.component, f.phase,

         COUNT(*)
         AS cells,

         SUM(c.state='AUTONOMOUS_VERIFIED')
         AS verified,
         SUM(c.state IN
              ('SKIP','NOT_APPLICABLE'))
              AS skipped,
         SUM(c.state IN
              ('REQUIRED','PENDING','AUTONOMOUS_DESIGNED','OPERATOR_ATTENDED_ONLY','FAIL'))
         AS open
  FROM features f JOIN coverage_cells c USING (feature_id)
  GROUP BY f.feature_id;
```

The closed-set skip_reason vocabulary (§11.4.3 / §11.4.6) is: single-node-selfhost (the DDOS/SCALE MVP defer), hardware_not_present (no real iOS device for the G3 MEM gate), topology_unsupported, feature_disabled_by_config, structurally-impossible (§11.4.112), phase_deferred (a re-arm cell, §10). A free-text reason outside this set is itself a gate finding.

4. The per-cell state machine

stateDiagram-v2

```
[*] --> REQUIRED: feature declares the type (§11.4.169)
REQUIRED --> PENDING: test authored, awaiting first run
PENDING --> AUTONOMOUS_DESIGNED: RED on broken artifact captured (§11.4.115)
AUTONOMOUS_DESIGNED --> AUTONOMOUS_VERIFIED: GREEN on CLEAN
deploy + evidence_path + artifact_md5 (§11.4.69/.108/.139)
AUTONOMOUS_VERIFIED --> FAIL: re-run regresses (§11.4.4 STOP-on-discovery)
FAIL --> AUTONOMOUS_DESIGNED: systematic-debug + fix (§11.4.102), re-RED, re-GREEN
REQUIRED --> OPERATOR_ATTENDED_ONLY: only a human-driven path exists (§11.4.52)
OPERATOR_ATTENDED_ONLY --> AUTONOMOUS_DESIGNED: autonomous path authored (promotion)
REQUIRED --> SKIP: topology/hardware absent (§11.4.3 closed reason)
REQUIRED --> NOT_APPLICABLE: type does not apply this phase (§11.4.6)
NOT_APPLICABLE --> REQUIRED: phase re-arm flips the cell (§10)
AUTONOMOUS_VERIFIED --> [*]: feature closeable when ALL declared cells VERIFIED or NA/SKIP-with-reason
```

A cell only reaches the terminal AUTONOMOUS_VERIFIED by passing through AUTONOMOUS_DESIGNED — i.e. a RED-on-the-broken-artifact reproduction MUST precede the GREEN (§11.4.115); a cell that jumps straight to AUTONOMOUS_VERIFIED without a recorded RED is the §11.4.43 "test added after the fix" bluff and is rejected by the generator (§7). The AUTONOMOUS_VERIFIED → FAIL → AUTONOMOUS_DESIGNED loop is the §11.4.4 STOP-on-discovery + §11.4.102 systematic-debug cycle made visible in the ledger.

5. The concrete HelixVPN feature inventory

The MVP feature set (the rows the ledger actually carries), grouped by component. Each row's evidence_class is its §11.4.69 sink-side class; transport rows trace to the [obf §2] survival matrix.

5.1 helix-core (Rust data plane) + transports

feature_id	title (≥40 chars)	evidence_class	decl type
F-TX-PLAIN-UDP	Plain-UDP transport carries WireGuard at ≥80% bare-link goodput	network_throughput	UNI INT BEN PER FA
F-TX-MASQUE-H3	MASQUE CONNECT-UDP (RFC 9298) transport survives a QUIC-SNI-fronted regime	network_throughput	UNI INT SEC BEN PER FA
F-TX-SHADOWSOCKS	Shadowsocks AEAD-TCP transport survives a hard UDP-block regime	network_throughput	UNI INT SEC
F-TX-UDP-OVER-TCP	UDP-over-TCP last-resort transport connects under UDP-block + TLS-SNI filter	network_throughput	UNI INT FA
F-TX-LW0	Lightweight WG-header obfuscation defeats a WG-fingerprint-only DPI regime	network_throughput	UNI INT SEC
F-TX-SELECT-LADDER	Regime-aware escalation ladder picks the surviving transport per detected regime	network_connectivity	UNI INT FA CHA
F-AUTHZ-REACH	Enrolled client reaches an authorized LAN host through the real tunnel	network_connectivity	UNI INT SEC STR CHA BEN

feature_id	title (≥40 chars)	evidence_class	declared types
F-DEFAULT-DENY	Unauthorized host is denied: SYN out, zero SYN-ACK, edge drop counter rises	network_connectivity	UI/F CHA UNI INT SEC CHA RAC CHA
F-KILLSWITCH	Kill-switch seals all egress on tunnel drop, zero plaintext + zero DNS leak	wifi_link (negative)	UNI E2E SEC CHA RAC UI/F CHA
F-DNS-LEAK	DNS queries forced through the tunnel resolver, none escape to the ISP resolver	wifi_link (negative)	UNI E2E SEC CHA CHA
F-IPV6-LEAK	IPv6 egress is routed-through-tunnel or blocked; no v6 escapes the seal	wifi_link (negative)	UNI E2E SEC
F-MTU-ENVELOPE	Each transport envelope's effective MTU is measured and Oversize is enforced	network_throughput	UNI PER
F-CORE-FSM	Client status FSM transitions Connecting→Connected→Reconnecting→Blocked correctly	counter delta	UNI RAC CON FA

5.2 helix-go (control plane) + helix-edge

feature_id	title (≥40 chars)	evidence_class	declared types
F-ENROLL	Device enrollment: client-generated keypair, private key never persists server-side	permission_grant	UNIT INT E2E SEC CONC FA
F-IPAM-ALLOC	IPAM allocates collision-free overlay ULA /48 addresses under concurrency	counter delta	UNIT INT CONC RACE
F-POLICY-COMPILE	Policy compiler turns ACL into AllowedIPs + verdict-map, fail-closed on parse error	network_connectivity	UNIT INT SEC

feature_id	title (≥40 chars)	evidence_class	declared types
F-POLICY-RECONCILE	Policy edit reconciles all devices in <1 s with no restart (SLO1)	counter delta	UNIT INT E2E STRESS CHAOS CONC RACE PERF CHAL
F-REVOKE	Device revoke removes the peer from every map and enforces at the edge in <1 s (SLO3)	counter delta	UNIT INT E2E SEC CHAOS CONC PERF CHAL
F-RLS-ISOLATION	Postgres RLS makes a tenant-B row invisible under a tenant-A query context	permission_grant (negative)	UNIT INT SEC
F-NO-LOG-SCHEMA	No durable per-connection/traffic/packet table exists; schema-lint enforces it	counter delta	UNIT INT SEC CHAL
F-EVENT-BACKBONE	Redis-Streams event → coordinator consume → minimal delta emitted, idempotent	counter delta	UNIT INT CHAOS CONC
F-WATCH-MAP	WatchNetworkMap server-stream emits snapshot + policy-filtered deltas to N streams	counter delta	UNIT INT E2E CONC RACE SCALE
F-EDGE-VERDICT	Edge verdict-map hot-swaps under live classification with no torn/half-applied map	network_connectivity	UNIT RACE STRESS
F-EDGE-PROBE	Edge returns decoy/silent/benign-RST to an unauthenticated active prober	network_connectivity (negative)	SEC FA
F-MTLS-CERT	Short-lived mTLS device cert gates the control stream; expired/revoked is refused	permission_grant	UNIT INT SEC

5.3 clients (Flutter UI + platform shims)

feature_id	title (≥40 chars)	evidence_class	declared types
F-UI-FLOWS	Three Flutter flavors (Access/Connector/Console) drive enroll→connect→reach end-to-end	REC	UI REC FA CHAL
F-FFI-BRIDGE	flutter_rust_bridge FFI lets Dart call connect() and receive the status stream	counter delta	UNIT UI FA
F-IOS-NE-MEM	iOS NEPacketTunnelProvider Rust core stays under the jetsam memory ceiling +30% headroom	MEM	MEM
F-ANDROID-VPN	Android VpnService + JNI builds the tun, survives background kill, RSS bounded	MEM	MEM E2E
F-SHIM-LINUX	Linux kernel-WG/tun shim integrates with systemd and the kill-switch state machine	network_connectivity	E2E SEC

F-IOS-NE-MEM declares only MEM because it is a platform-resource constraint, not a reachability feature; its cell is SKIP: hardware_not_present on any host without a real iOS device (§11.4.3 — never a simulator PASS, [10 QA-D3]).

5.4 Phase-2 parked features (MVP NOT_APPLICABLE)

feature_id	title (≥40 chars)	evidence_class	MVP state
F-DDOS-FLOOD	Handshake-flood + volumetric flood: edge rate-limits and fails static, never crashes	network_throughput	NOT_APPLICABLE: single-node-selfhost
F-SCALE-HA	N simulated agents hold WatchNetworkMap streams; convergence + bounded memory / 24 h	MEM	NOT_APPLICABLE: phase_deferred (partial SLO4 soak at MVP)
F-DAITA-SHAPE	DAITA size/timing/cover-traffic defense (maybe not) flattens the traffic-analysis signature	network_throughput	NOT_APPLICABLE: phase_deferred

6. The full ledger table sketch (every MVP feature × type)

The rendered docs/qa/coverage_ledger.md table (✓ = AUTONOMOUS_VERIFIED, ■ = authored/DESIGNED, — = no cell, NA = NOT_APPLICABLE, S = SKIP, AC = bound DoD Challenge). **This sketch is the illustrative *target* state at the MVP gate — NOT the current state.** At spec time no test has run, so every ✓ shown here is schema-impossible *today*: the §2/§3 CHECK forbids AUTONOMOUS_VERIFIED without a captured evidence_path + artifact_md5. The generator (§7) therefore seeds every declared cell at **REQUIRED** (or **PENDING** once a test is authored) and advances a cell to ✓ only when a real clean-deploy run lands its evidence; the live values are always generated from the DB (§7), never hand-set to this target.

feature_id	UNIT	INT	E2E	SEC	STRESS	CHAOS	CONC	RACE	MEM
F-TX-PLAIN-UDP	✓	✓	✓	—	—	—	—	—	—
F-TX-MASQUE-H3	✓	✓	✓	✓	—	—	—	—	—
F-TX-SHADOWSOCKS	✓	✓	✓	✓	—	—	—	—	—
F-TX-UDP-OVER-TCP	✓	✓	✓	—	—	—	—	—	—
F-TX-LWO	✓	✓	✓	✓	—	—	—	—	—
F-TX-SELECT-LADDER	✓	✓	✓	—	—	✓	—	—	—
F-AUTHZ-REACH	✓	✓	✓	✓	✓	✓	—	—	—
F-DEFAULT-DENY	✓	✓	✓	✓	—	✓	—	✓	—
F-KILLSWITCH	✓	—	✓	✓	—	✓	—	✓	—
F-DNS-LEAK	✓	—	✓	✓	—	✓	—	—	—
F-IPV6-LEAK	✓	—	✓	✓	—	—	—	—	—
F-CORE-FSM	✓	—	—	—	—	—	✓	✓	—
F-ENROLL	✓	✓	✓	✓	—	—	✓	—	—
F-IPAM-ALLOC	✓	✓	—	—	—	—	✓	✓	—
F-POLICY-RECONCILE	✓	✓	✓	—	✓	✓	✓	✓	—
F-REVOKE	✓	✓	✓	✓	—	✓	✓	—	—
F-RLS-ISOLATION	✓	✓	—	✓	—	—	—	—	—
F-NO-LOG-SCHEMA	✓	✓	—	✓	—	—	—	—	—
F-WATCH-MAP	✓	✓	✓	—	—	—	✓	✓	—

feature_id	UNIT	INT	E2E	SEC	STRESS	CHAOS	CONC	RACE	MEM
F-EDGE-PROBE	—	—	—	✓	—	—	—	—	—
F-MTLS-CERT	✓	✓	—	✓	—	—	—	—	—
F-UI-FLOWS	—	—	—	—	—	—	—	—	—
F-IOS-NE-MEM	—	—	—	—	—	—	—	—	✓/S ¹
F-ANDROID-VPN	—	—	✓	—	—	—	—	—	✓
F-DDOS-FLOOD	NA ²	NA ²	—	NA ²	—	—	—	—	—
F-SCALE-HA	—	—	—	—	—	—	—	—	NA ³

¹ MEM cell is AUTONOMOUS_VERIFIED on a host with a real iOS device, else SKIP: hardware_not_present ([10 QA-D3]). ² NOT_APPLICABLE: single-node-selfhost at MVP; re-arms Phase 2 ([10 §5.8]). ³ NOT_APPLICABLE: phase_deferred; partial SLO4 24 h soak runs at MVP, full SCALE Phase 2.

7. Generation + sync pipeline (coverage_ledger tool + docs_chain §11.4.106)

The ledger is **generated, never hand-edited** — hand-editing a cell would let a human assert a state the evidence does not back (the exact bluff the schema CHECKs forbid). The pipeline:

```

flowchart LR
    SPEC[spec feature map<br/>§5 inventory + FR/NFR refs] -->
    GEN[coverage_ledger tool<br/>tools/coverage_ledger/]
    HQA[helix_qa events.jsonl<br/>§11.4.116 verdict+evidence] -->
    GEN
    DIARY[test_diary rows<br/>§11.4.149] --> GEN
    RSIG[runtime-signature registry<br/>§11.4.108] --> GEN
    GEN --> DB[(docs/.workable_items.db<br/>coverage_cells
    §11.4.95)]
    DB --> DC[docs_chain sync<br/>.docs_chain/contexts/
    helixvpn_coverage.yaml]
    DC --> MD[docs/qa/coverage_ledger.md]
    DC --> HTML[coverage_ledger.html]
    DC --> PDF[coverage_ledger.pdf]
```

The generator contract (tools/coverage_ledger, Go, deterministic, minimal-LLM per §11.4.141 — pure data transform):

1. Read the §5 feature inventory + the §11.4.169 declared-type set per feature.
2. For each (feature, type) cell, read the **most recent** helix_qa verdict event (§11.4.116 — the event carries the evidence path; a PASS event with no evidence path is a contradiction → the generator records FAIL, never AUTONOMOUS_VERIFIED, defeating B5 at the ledger layer).

3. Cross-check the verdict's `artifact_md5` against the **current clean-deploy** artifact (§11.4.139) — a verdict from a stale artifact downgrades the cell to `PENDING` (defeats B4 stale-state, [10 §0]).
4. Enforce the **RED-precedes-GREEN** invariant: a cell may be `AUTONOMOUS_VERIFIED` only if the diary holds a prior `RED_MODE=1` reproduction row for it (§11.4.115/.43) — else the generator records `AUTONOMOUS_DESIGNED` and emits a finding.
5. Enforce determinism: a `*_VERIFIED` cell records `n_iters ≥ 3` (normal) / `≥ 10` (cycle-validation) with identical evidence-hashes (§11.4.50); a divergent run forces `FAIL`.
6. Write the `coverage_cells` rows, WAL-checkpoint, and hand off to `docs_chain`.

docs_chain sync (§11.4.106). The `.docs_chain/contexts/helixvpn_coverage.yaml` context declares the `DB→Markdown→HTML→PDF` chain; `docs_chain sync` regenerates the rendered ledger with content-hash change detection (§11.4.86, not `mtime`) and an atomic-rename commit; `docs_chain verify` is the deterministic CI/pre-build gate (`make docs-verify`, [10 §9]) that FAILs if the rendered ledger drifts from the DB. A missing `pandoc/weasyprint` surfaces a typed `ToolAbsentError` + honest §11.4.3 `SKIP`, never a fake export (§11.4.106).

```
# .docs_chain/contexts/helixvpn_coverage.yaml (§11.4.106 –
# consumer-owned context)
context: helixvpn-coverage-ledger
source:
  kind: sqlite
  path: docs/.workable_items.db
  query_view: v_feature_rollup           # + coverage_cells join,
    rendered to a matrix
derived:
  - { kind: markdown, path: docs/qa/coverage_ledger.md }
  - { kind: html,     path: docs/qa/coverage_ledger.html } #
    §11.4.65
  - { kind: pdf,      path: docs/qa/coverage_ledger.pdf }
change_detection: content_hash           # §11.4.86 drift-proof
    (NOT mtime)
on_tool_absent: skip_with_reason         # §11.4.106 – never a
    fake transform
```

The ledger generation runs on every `make qa / make coverage` and at the §11.4.40 pre-tag sweep, backgrounded (§11.4.89) so the main work stream is never blocked ([10 §9]).

8. The release-gate rule (what blocks a tag)

The single mechanical release rule ([10 §6.2], made precise):

A phase `P` ships **only when** `SELECT count(*) FROM v_coverage_gaps WHERE phase='P'` returns `0` — i.e. every declared cell for that phase is `AUTONOMOUS_VERIFIED`, or `NOT_APPLICABLE/SKIP` with a closed-set reason.

Consequences, each a §11.4.169 / §11.4.52 / §11.4.40 obligation:

- A cell in REQUIRED / PENDING / AUTONOMOUS_DESIGNED is **unfinished work** — it blocks the tag (the feature declared the type but has no captured GREEN).
- A cell in OPERATOR_ATTENDED_ONLY blocks the tag until promoted, citing a tracked migration item (§11.4.52) — honest, but it does not scale to the §11.4.156 local-re-run regime.
- A cell in FAIL blocks the tag and triggers the §11.4.4 STOP + §11.4.102 systematic-debug + §11.4.129 huge-blocker restart if release-blocking-severity.
- SKIP / NOT_APPLICABLE do **not** block — but the closed-set reason is audited every §11.4.40 / §11.4.42 sweep (a SKIP: hardware_not_present that persists past the phase where the hardware should exist is itself a finding).

The ledger is the §11.4.40 full-suite-retest scoreboard: the retest run *is* what flips cells to AUTONOMOUS_VERIFIED; the tag is cut only when the scoreboard is clean for the phase. The §11.4.126 endless-loop terminal condition (a published, prefixed tag, §11.4.151) is therefore **mechanically** v_coverage_gaps empty + the §7.2 ACs all CHAL-scored PASS ([10 §7.2]).

9. Anti-bluff: the §1.1 paired mutations

The ledger's own gates are anti-bluff per §1.1 — each gate ships with a paired mutation that **MUST** make it FAIL:

Gate	What it asserts	Paired §1.1 mutation → expected FAIL
CM-LEDGER-EVIDENCE-REQUIRED	no AUTONOMOUS_VERIFIED cell without an evidence_path	seed a VERIFIED cell with evidence_path=NULL → the schema CHECK rejects the INSERT → mutation caught
CM-LEDGER-NO-SILENT-SKIP	every SKIP/NA cell carries a closed-set reason	seed a SKIP cell with skip_reason=NULL → CHECK rejects → caught
CM-LEDGER-RED-PRECEDES-GREEN	a VERIFIED cell has a prior diary RED row (§11.4.115)	mark a cell VERIFIED with no RED diary row → generator records AUTONOMOUS_DESIGNED + finding → gate FAILs on the forged cell
CM-LEDGER-NO-STALE-ARTIFACT	a VERIFIED cell's artifact_md5 == current clean deploy (§11.4.139)	inject a verdict from a stale artifact_md5 → generator downgrades to PENDING → the forged "verified" is gone
CM-LEDGER-GAP-BLOCKS-TAG	a non-empty v_coverage_gaps blocks the tag	inject an OPERATOR_ATTENDED_ONLY cell into the release phase → the pre-tag gate refuses → caught
CM-LEDGER-DETERMINISTIC	a VERIFIED cell has n_iters ≥ 3 identical (§11.4.50)	seed a cell whose two run-hashes diverge → generator forces FAIL → the green is gone

The deepest anti-bluff property is **structural, not procedural**: because `AUTONOMOUS_VERIFIED` is unrepresentable in the schema without a non-empty `evidence_path + artifact_md5`, the most common ledger bluff (marking a feature "covered" with no captured artifact) cannot be written to the database at all — the `INSERT/UPDATE` fails the `CHECK`. The ledger inherits the [10 §0] B1–B5 defences: B1 (config-only) and B2 (absence-of-error) are blocked by the evidence-path `CHECK`; B4 (stale-state) by the `artifact_md5` cross-check; B5 (unvalidated analyzer) by the generator treating an evidence-less `PASS` event as `FAIL`.

10. Phase re-arm: cells that flip across phases

A `NOT_APPLICABLE/phase_deferred` cell is not dead — it is **parked** and re-arms mechanically when its phase opens ([10 §7.3]). The generator re-reads the per-feature declared-type set each run; when a feature's phase advances (e.g. `F-DDOS-FLOOD` from `MVP-NA` to `Phase-2-REQUIRED`), the matching cells flip `NOT_APPLICABLE` → `REQUIRED` and immediately appear in `v_coverage_gaps` for the new phase — so the Phase-2 DDoS/SCALE suite cannot be silently forgotten (§11.4.118):

- **Phase 2 re-arms**: `F-DDOS-FLOOD` (DDoS), `F-SCALE-HA` (full SCALE + WatchNetworkMap 10k-stream soak), `F-DAITA-SHAPE` (maybenot traffic-analysis defense, [research-daita_test §4]), plus P2P NAT-traversal + multi-hop nested-WG + PQ-handshake-interop E2E rows.
- **Phase 3 re-arms**: HarmonyOS NEXT + Aurora OS shim E2E/MEM rows, WASM browser-MASQUE E2E, and a third-party-audit + reproducible-build attestation row ([10 §7.3]).

Each re-arm is a *visible* state transition in the ledger, never an edit — the parked cell with its `phase_deferred` reason is the auditable record that the work is owed, not omitted.

11. Edge cases & honest gaps

#	Edge case	Required behavior	Cite
L1	A feature exists in code but has no features row	reconciliation gap → the §11.4.153 feature-status audit FAILs (a code-present feature missing from the ledger is a §11.4.118 bluff)	§0, §11.4.153
L2	A features row with no backing code	dead-row → investigate per §11.4.124 before removing; never silent-delete	§11.4.124
L3	A <code>VERIFIED</code> cell whose <code>evidence_path</code> file was deleted	<code>docs_chain verify</code> + a path-existence check downgrade it to <code>PENDING</code> (evidence must be re-captured)	§7, §11.4.69
L4	Two QA runs disagree on a cell		§7 step 3

#	Edge case	Required behavior	Cite
		the more-recent run wins only if its artifact_md5 is the current clean deploy; else the cell is PENDING	
L5	An OPERATOR_ATTENDED_ONLY cell with no migration item	itself a finding — §11.4.52 requires the tracked promotion item	§8
L6	SKIP: hardware_not_present on the only host that has the hardware	the SKIP is dishonest there → the §11.4.40 sweep audits SKIP reasons against the running host's capability	§8

Honest boundary (§11.4.6). The ledger guarantees *every enumerated feature × its warranted types is proven with cited evidence on a clean artifact* — it does **not** prove the absence of un-enumerated defects (the §11.4.118 unknown-unknown surface, which the discovery-pressure pass addresses), and a 100%-green ledger with an incomplete §5 inventory is a §11.4.153 completeness bluff, not a coverage proof. The inventory's completeness is itself audited by the §11.4.153 per-feature Status-document reconciliation against the codebase. **UNVERIFIED:** the exact final MVP feature count — §5 is the spec-stage inventory; the live count is whatever the §11.4.153 audit reconciles against the implemented codebase.

Sources verified

- [10] docs/research/mvp/final/10-testing-acceptance-and-qa.md §2 (test-type taxonomy), §3 (evidence model), §6 (ledger DDL sketch + state machine + release rule), §7 (acceptance gates / ACs / SLOs), §9 (local gate harness) — read 2026-06-26.
- [obf] docs/research/mvp/final/v02-data-plane/obfuscation-and-dpi.md §2 (regime → transport survival matrix the transport-feature rows trace to), §7 (the T-id test points the CHAL cells bind) — read 2026-06-26.
- [research-daita_test] docs/research/mvp/final/v09-research/research-daita_test.md §4 (DAITA v2 — the parked F-DAITA-SHAPE Phase-2 row) — read 2026-06-26.
- [MASTER_INDEX] docs/research/mvp/final/MASTER_INDEX.md (Volume 8 table, this doc's slot) — read 2026-06-26.
- Constitution: §11.4.25 (full-automation coverage ledger), §11.4.52 (autonomous classification + OPERATOR_ATTENDED_ONLY release-blocker), §11.4.69 (sink-side evidence taxonomy + evidence-path requirement), §11.4.93/95 (git-tracked workable-items SQLite SSoT), §11.4.106 (docs_chain), §11.4.108/139 (runtime signature / clean-artifact), §11.4.115/43 (RED-precedes-GREEN), §11.4.50 (determinism), §11.4.118 (discovery-pressure / no silent gap), §11.4.153 (per-feature status reconciliation), §11.4.156 (no remote CI), §11.4.169 (test-type mandate) — CLAUDE.md, read 2026-06-26.